Stueckelberg Trick

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1 Stueckelberg Trick

A mass term which breaks gauge symmetry can be rewritten with a compensator field into a gauge-invariant form.

1.1 Gauge Boson + Mass: Why is it a Problem?

Consider an Abelian gauge field A_μ (like the photon).

1.1.1 Maxwell Theory

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu}, \qquad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$$

This theory has gauge symmetry:

$$A_\mu \to A_\mu + \partial_\mu \alpha$$

Consequences:

- Only 2 physical polarizations
- Renormalizability

1.1.2 Proca Theory: Add a Mass

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}m^2 A_\mu A^\mu$$

Now gauge invariance is lost, since under gauge transformation:

$$A_\mu A^\mu$$

is not invariant.
Degrees of freedom:

- Massless vector $\to$ 2 transverse d.o.f.
- Massive vector $\to$ 2 transverse $+1$ longitudinal $= 3$ d.o.f.

1.2 Stueckelberg Trick

Rewrite the longitudinal mode as a scalar field.

Introduce a scalar field:

$$\phi(x)$$

Replace:

$$A_\mu \to A_\mu - \frac{1}{m}\partial_\mu \phi$$

Then the Lagrangian becomes:

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}m^2 \left(A_\mu - \frac{1}{m}\partial_\mu \phi \right)^2$$

This is gauge invariant if we define:

$$A_\mu \to A_\mu + \partial_\mu \alpha$$

$$\phi \to \phi + m\alpha$$

Then:

$$A_\mu - \frac{1}{m}\partial_\mu \phi \to A_\mu + \partial_\mu \alpha - \frac{1}{m}\partial_\mu(\phi + m\alpha)$$

$$= A_\mu + \partial_\mu \alpha - \frac{1}{m}\partial_\mu \phi - \partial_\mu \alpha = A_\mu - \frac{1}{m}\partial_\mu \phi$$

So the combination is unchanged.

Therefore the mass term is now gauge invariant.

2 Illustration of Degrees of Freedom

2.1 Massless Spin-1 Particle in $R^{3,1}$

A massless spin-1 particle (photon) has:

$$2 physical degrees of freedom$$

corresponding to helicities:

$$\lambda = \pm 1$$

These are the two transverse polarizations.

2.2 Vector Field Components vs Physical Degrees of Freedom

A vector field has four components:

$$A_\mu = (A_0, A_1, A_2, A_3)$$

However:

$$field components \neq physical degrees of freedom$$

because constraints, gauge redundancy, and equations of motion reduce the number of physical modes.

2.3 Maxwell Theory Counting

Start with:

$$4 components$$

Gauge freedom removes one:

$$-1$$

Gauge constraint removes one more:

$$-1$$

Thus:

$$4 - 1 - 1 = 2$$

So Maxwell theory has:

$$2 physical d.o.f.$$

2.4 Photon Moving in the z-Direction

Assume the photon momentum is:

$$k^\mu = (\omega, 0, 0, \omega)$$

This means the wave propagates along the z-axis.
Then physical oscillations occur only in the x and y directions.
The two transverse polarization vectors are:

$$\epsilon^\mu_{(1)} = (0, 1, 0, 0)$$

$$\epsilon^\mu_{(2)} = (0, 0, 1, 0)$$

2.5 Plane Wave Solution

Take:

$$A^\mu(x) = \epsilon^\mu e^{-ik\cdot x}$$

where:

- $k^\mu = (\omega, \vec{k})$ is the four-momentum
- ϵ^μ is the polarization vector

Then:

$$F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu = -i(k^\mu \epsilon^\nu - k^\nu \epsilon^\mu)e^{-ikx}$$

2.6 Maxwell Equation

The vacuum Maxwell equation is:

$$\partial_\mu F^{\mu\nu} = 0$$

Substituting the plane wave gives:

$$k^2 \epsilon^\nu - k^\nu (k \cdot \epsilon) = 0$$

2.7 Massless Condition

For a massless photon:

$$k^\mu k_\mu = 0$$

so:

$$k^2 = 0$$

Therefore:

$$k^\nu (k \cdot \epsilon) = 0$$

Since $k^\nu \neq 0$:

$$k \cdot \epsilon = 0$$

This is the transversality condition.

2.8 For Motion Along z

Using:

$$k^\mu = (\omega, 0, 0, \omega)$$

we get:

$$\omega\epsilon^0 - \omega\epsilon^3 = 0$$

Hence:

$$\epsilon^0 = \epsilon^3$$

This is the Gauss constraint arising from Maxwell equations.

3 Gauge Freedom and Massive Vector Polarizations

3.1 Gauge Freedom

For a massless photon moving in the z-direction, the polarization vector may be written as

$$\epsilon^\mu = (a, b, c, a)$$

where:

- b, c are transverse components (x, y directions)
- a corresponds to time / longitudinal part

3.2 Electromagnetic Gauge Symmetry

For the electromagnetic field:

$$A_\mu \to A_\mu + \partial_\mu \lambda$$

Take a plane-wave gauge parameter:

$$\lambda(x) = \lambda\, e^{-ik\cdot x}$$

Then:

$$\partial_\mu \lambda = -ik_\mu \lambda e^{-ikx}$$

So the gauge transformation shifts:

$$A'^\mu(x) = (\epsilon^\mu + \lambda k^\mu)\, e^{-ikx}$$

Thus:

$$\epsilon^\mu \sim \epsilon^\mu + \lambda k^\mu$$

Both describe the same photon state.
Hence polarization vectors are defined only up to:

$$[\epsilon^\mu]$$

the same gauge equivalence class.

3.3 Gauge Choice Removes Longitudinal Mode

Gauge transformations change the time component and longitudinal component because the shift is along k^μ.

For momentum:

$$k^\mu = (\omega, 0, 0, \omega)$$

choose λ so that $a = 0$, giving:

$$\epsilon^\mu = (0, b, c, 0)$$

Therefore only transverse modes remain.
Equivalently:

$$\vec{\epsilon} \cdot \vec{k} = 0$$

so polarization is perpendicular to propagation direction.

4 Massive Vector Field

The Proca Lagrangian is:

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}m^2 A_\mu A^\mu$$

Since the mass term breaks gauge symmetry, we can no longer gauge away one component.

4.1 Degree of Freedom Counting

We still have one constraint from equations of motion, so:

$$4 - 1 = 3$$

Thus a massive spin-1 particle has three physical polarizations.

5 Polarization States of Massive Vector

Assume the particle moves in the z-direction.

Two transverse modes remain unchanged:

$$\epsilon_1^\mu = (0, 1, 0, 0)$$

$$\epsilon_2^\mu = (0, 0, 1, 0)$$

There is also one longitudinal mode.

In the rest frame:

$$p^\mu = (m, 0, 0, 0)$$

Boosting to momentum along z gives:

$$\epsilon_L^\mu \propto (p, 0, 0, E)$$

which is the longitudinal polarization.

6 Group Theory Interpretation

A massive spin-1 particle has a rest frame, so states transform under:

$$SO(3)$$

giving three spin states:

$$m_s = -1, 0, +1$$

A massless vector has:

- gauge symmetry
- no rest frame

Therefore it has only two helicity states:

$$\lambda = \pm 1$$